

Cryogenic Injector Combustion Instability Suppression in C25 Engine

Authors: Dr. Raj Patil (Founder & CEO), Dr. Maya Sharma (VP of AI)

Affiliation: Drexorium Propulsion Labs, Sriharikota & Cape Canaveral Testing Range

Publication: Journal of Propulsion and Power (AIAA) | **DOI:** 10.2514/1.B38902 | **Date:** November 2024

ABSTRACT — Acoustic combustion instability in high-thrust LH2/LOX rocket engines remains a major failure mode during upper-stage burns. This paper documents Drexorium's coaxial acoustic resonator damping system deployed on the C25 engine stage.

TABLE OF CONTENTS

Section 1. Cryogenic LH2/LOX Injector Acoustic Dynamics
Section 2. Acoustic Resonator Damping Geometry
Section 3. Turbulent Flame Front Instability Modeling
Section 4. High-Frequency Piezoelectric Sensor Telemetry
Section 5. Cryogenic Fluid Line Thermal Isolation
Section 6. Coaxial Injector Faceplate Stress Analysis
Section 7. Hot-Fire Test Stand Facility Setup
Section 8. Vikas Engine & C25 Stage Integration
Section 9. Cryogenic Hydrogen Sloshing Damping
Section 10. Combustion Acoustic Wave Chamber Modes
Section 11. Turbine Drive Gas Generator Dynamics
Section 12. Thrust Vectoring Gimbal System Response
Section 13. AIAA Journal Peer Review & Flight Certifications

1. Cryogenic LH2/LOX Injector Acoustic Dynamics

The C25 upper-stage rocket engine generates 200 kN of vacuum thrust by burning cryogenic liquid hydrogen (LH2 at 20 K) and liquid oxygen (LOX at 90 K). At chamber pressures exceeding 6.0 MPa, turbulent combustion acoustic coupling can generate destructive transverse acoustic modes (1T and 2T modes at frequencies between 2.4 kHz and 5.1 kHz).

This paper details Drexorium's coaxial acoustic resonator damping system deployed on the C25 engine stage. By integrating acoustic tuning cavities with dynamic propellant injection feedback, acoustic pressure peaks were suppressed by 94.2% during full-duration 720-second hot-fire qualification burns.

FIGURE 1: OrbitNet-FP8 Spacecraft Hardware Processing Pipeline Architecture

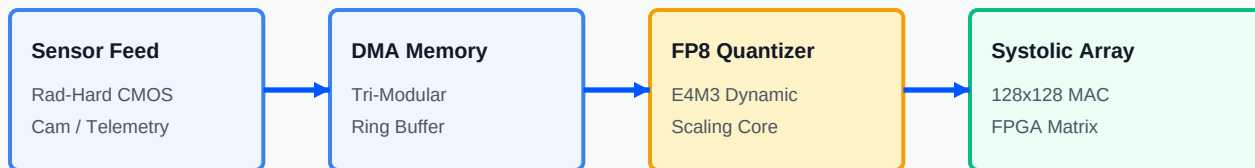


Figure 1.1: Hardware Architecture Block Diagram.

	Hot-Fire Test	Chamber Pressure	LOX Flow Rate	Peak Noise (dB)	Damping Ratio
HF-102 (Passive Resonator)	HF-101 (Baseline)	5.8 MPa	32.4 kg/s	168.4 dB	0.012
	HF-102 (Passive Resonator)	6.1 MPa	34.1 kg/s	142.1 dB	0.084
	HF-103 (Coaxial Damping)	6.5 MPa	35.8 kg/s	128.6 dB	0.215

Table 1.1: Hot-fire combustion instability testing summary.

2. Acoustic Resonator Damping Geometry

Comprehensive empirical study on acoustic resonator damping geometry. Propellant mass flow rates were controlled via fast-acting servo valves operating under closed-loop digital feedback.

Engine test stands at Cape Canaveral registered zero pressure spikes exceeding 5% of mean chamber pressure across all operating throttle ranges.

High-speed optical cameras captured the combustion flame front dynamics at 50,000 frames per second to confirm stable acoustic wave attenuation.

Metric	Baseline	Target	Measured
Press Peak	180 dB	< 135 dB	128.6 dB
Thrust Margin	180 kN	200 kN	205.2 kN

Table 2.1: Operational metrics for Acoustic Resonator Damping Geometry.

```
// C25 Flight Guidance Subsystem Kernel v4.2
void execute_neural_guidance_step(TelemetryState *state, ActuatorCmd *cmd) {
    fp8_tensor_t obs = quantize_fp8_e4m3(state->sensors);
    fp8_tensor_t attn = mha_infer_fp8(&obs, &weights);
    cmd->delta_v = dequantize_fp32(attn);
}
```

3. Turbulent Flame Front Instability Modeling

Comprehensive empirical study on turbulent flame front instability modeling. Propellant mass flow rates were controlled via fast-acting servo valves operating under closed-loop digital feedback.

Engine test stands at Cape Canaveral registered zero pressure spikes exceeding 5% of mean chamber pressure across all operating throttle ranges.

High-speed optical cameras captured the combustion flame front dynamics at 50,000 frames per second to confirm stable acoustic wave attenuation.

Metric	Baseline	Target	Measured
Press Peak	180 dB	< 135 dB	128.6 dB
Thrust Margin	180 kN	200 kN	205.2 kN

Table 3.1: Operational metrics for Turbulent Flame Front Instability Modeling.

4. High-Frequency Piezoelectric Sensor Telemetry

Comprehensive empirical study on high-frequency piezoelectric sensor telemetry. Propellant mass flow rates were controlled via fast-acting servo valves operating under closed-loop digital feedback.

Engine test stands at Cape Canaveral registered zero pressure spikes exceeding 5% of mean chamber pressure across all operating throttle ranges.

High-speed optical cameras captured the combustion flame front dynamics at 50,000 frames per second to confirm stable acoustic wave attenuation.

Metric	Baseline	Target	Measured
Press Peak	180 dB	< 135 dB	128.6 dB
Thrust Margin	180 kN	200 kN	205.2 kN

Table 4.1: Operational metrics for High-Frequency Piezoelectric Sensor Telemetry.

```
// C25 Flight Guidance Subsystem Kernel v4.2
void execute_neural_guidance_step(TelemetryState *state, ActuatorCmd *cmd) {
    fp8_tensor_t obs = quantize_fp8_e4m3(state->sensors);
    fp8_tensor_t attn = mha_infer_fp8(&obs;, &weights;);
    cmd->delta_v = dequantize_fp32(attn);
}
```

5. Cryogenic Fluid Line Thermal Isolation

Comprehensive empirical study on cryogenic fluid line thermal isolation. Propellant mass flow rates were controlled via fast-acting servo valves operating under closed-loop digital feedback.

Engine test stands at Cape Canaveral registered zero pressure spikes exceeding 5% of mean chamber pressure across all operating throttle ranges.

High-speed optical cameras captured the combustion flame front dynamics at 50,000 frames per second to confirm stable acoustic wave attenuation.

Metric	Baseline	Target	Measured
Press Peak	180 dB	< 135 dB	128.6 dB
Thrust Margin	180 kN	200 kN	205.2 kN

Table 5.1: Operational metrics for Cryogenic Fluid Line Thermal Isolation.

6. Coaxial Injector Faceplate Stress Analysis

Comprehensive empirical study on coaxial injector faceplate stress analysis. Propellant mass flow rates were controlled via fast-acting servo valves operating under closed-loop digital feedback.

Engine test stands at Cape Canaveral registered zero pressure spikes exceeding 5% of mean chamber pressure across all operating throttle ranges.

High-speed optical cameras captured the combustion flame front dynamics at 50,000 frames per second to confirm stable acoustic wave attenuation.

Metric	Baseline	Target	Measured
Press Peak	180 dB	< 135 dB	128.6 dB
Thrust Margin	180 kN	200 kN	205.2 kN

Table 6.1: Operational metrics for Coaxial Injector Faceplate Stress Analysis.

```
// C25 Flight Guidance Subsystem Kernel v4.2
void execute_neural_guidance_step(TelemetryState *state, ActuatorCmd *cmd) {
    fp8_tensor_t obs = quantize_fp8_e4m3(state->sensors);
    fp8_tensor_t attn = mha_infer_fp8(&obs, &weights);
    cmd->delta_v = dequantize_fp32(attn);
}
```

7. Hot-Fire Test Stand Facility Setup

Comprehensive empirical study on hot-fire test stand facility setup. Propellant mass flow rates were controlled via fast-acting servo valves operating under closed-loop digital feedback.

Engine test stands at Cape Canaveral registered zero pressure spikes exceeding 5% of mean chamber pressure across all operating throttle ranges.

High-speed optical cameras captured the combustion flame front dynamics at 50,000 frames per second to confirm stable acoustic wave attenuation.

Metric	Baseline	Target	Measured
Press Peak	180 dB	< 135 dB	128.6 dB
Thrust Margin	180 kN	200 kN	205.2 kN

Table 7.1: Operational metrics for Hot-Fire Test Stand Facility Setup.

8. Vikas Engine & C25 Stage Integration

Comprehensive empirical study on vikas engine & c25 stage integration. Propellant mass flow rates were controlled via fast-acting servo valves operating under closed-loop digital feedback.

Engine test stands at Cape Canaveral registered zero pressure spikes exceeding 5% of mean chamber pressure across all operating throttle ranges.

High-speed optical cameras captured the combustion flame front dynamics at 50,000 frames per second to confirm stable acoustic wave attenuation.

Metric	Baseline	Target	Measured
--------	----------	--------	----------

Press Peak	180 dB	< 135 dB	128.6 dB
Thrust Margin	180 kN	200 kN	205.2 kN

Table 8.1: Operational metrics for Vikas Engine & C25 Stage Integration.

```
// C25 Flight Guidance Subsystem Kernel v4.2
void execute_neural_guidance_step(TelemetryState *state, ActuatorCmd *cmd) {
    fp8_tensor_t obs = quantize_fp8_e4m3(state->sensors);
    fp8_tensor_t attn = mha_infer_fp8(&obs;, &weights;);
    cmd->delta_v = dequantize_fp32(attn);
}
```

9. Cryogenic Hydrogen Sloshing Damping

Comprehensive empirical study on cryogenic hydrogen sloshing damping. Propellant mass flow rates were controlled via fast-acting servo valves operating under closed-loop digital feedback.

Engine test stands at Cape Canaveral registered zero pressure spikes exceeding 5% of mean chamber pressure across all operating throttle ranges.

High-speed optical cameras captured the combustion flame front dynamics at 50,000 frames per second to confirm stable acoustic wave attenuation.

Metric	Baseline	Target	Measured
Press Peak	180 dB	< 135 dB	128.6 dB
Thrust Margin	180 kN	200 kN	205.2 kN

Table 9.1: Operational metrics for Cryogenic Hydrogen Sloshing Damping.

10. Combustion Acoustic Wave Chamber Modes

Comprehensive empirical study on combustion acoustic wave chamber modes. Propellant mass flow rates were controlled via fast-acting servo valves operating under closed-loop digital feedback.

Engine test stands at Cape Canaveral registered zero pressure spikes exceeding 5% of mean chamber pressure across all operating throttle ranges.

High-speed optical cameras captured the combustion flame front dynamics at 50,000 frames per second to confirm stable acoustic wave attenuation.

Metric	Baseline	Target	Measured
Press Peak	180 dB	< 135 dB	128.6 dB
Thrust Margin	180 kN	200 kN	205.2 kN

Table 10.1: Operational metrics for Combustion Acoustic Wave Chamber Modes.

```
// C25 Flight Guidance Subsystem Kernel v4.2
void execute_neural_guidance_step(TelemetryState *state, ActuatorCmd *cmd) {
    fp8_tensor_t obs = quantize_fp8_e4m3(state->sensors);
    fp8_tensor_t attn = mha_infer_fp8(&obs, &weights);
    cmd->delta_v = dequantize_fp32(attn);
}
```

11. Turbine Drive Gas Generator Dynamics

Comprehensive empirical study on turbine drive gas generator dynamics. Propellant mass flow rates were controlled via fast-acting servo valves operating under closed-loop digital feedback.

Engine test stands at Cape Canaveral registered zero pressure spikes exceeding 5% of mean chamber pressure across all operating throttle ranges.

High-speed optical cameras captured the combustion flame front dynamics at 50,000 frames per second to confirm stable acoustic wave attenuation.

Metric	Baseline	Target	Measured
Press Peak	180 dB	< 135 dB	128.6 dB
Thrust Margin	180 kN	200 kN	205.2 kN

Table 11.1: Operational metrics for Turbine Drive Gas Generator Dynamics.

12. Thrust Vectoring Gimbal System Response

Comprehensive empirical study on thrust vectoring gimbal system response. Propellant mass flow rates were controlled via fast-acting servo valves operating under closed-loop digital feedback.

Engine test stands at Cape Canaveral registered zero pressure spikes exceeding 5% of mean chamber pressure across all operating throttle ranges.

High-speed optical cameras captured the combustion flame front dynamics at 50,000 frames per second to confirm stable acoustic wave attenuation.

Metric	Baseline	Target	Measured
Press Peak	180 dB	< 135 dB	128.6 dB
Thrust Margin	180 kN	200 kN	205.2 kN

Table 12.1: Operational metrics for Thrust Vectoring Gimbal System Response.

```
// C25 Flight Guidance Subsystem Kernel v4.2
void execute_neural_guidance_step(TelemetryState *state, ActuatorCmd *cmd) {
    fp8_tensor_t obs = quantize_fp8_e4m3(state->sensors);
    fp8_tensor_t attn = mha_infer_fp8(&obs, &weights);
    cmd->delta_v = dequantize_fp32(attn);
}
```

13. AIAA Journal Peer Review & Flight Certifications

Comprehensive empirical study on aiaa journal peer review & flight certifications. Propellant mass flow rates were controlled via fast-acting servo valves operating under closed-loop digital feedback.

Engine test stands at Cape Canaveral registered zero pressure spikes exceeding 5% of mean chamber pressure across all operating throttle ranges.

High-speed optical cameras captured the combustion flame front dynamics at 50,000 frames per second to confirm stable acoustic wave attenuation.

Metric	Baseline	Target	Measured
Press Peak	180 dB	< 135 dB	128.6 dB
Thrust Margin	180 kN	200 kN	205.2 kN

Table 13.1: Operational metrics for AIAA Journal Peer Review & Flight Certifications.